scoreboard.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/env) (5 of 5) - GVIM (on ip-10-115-24-57.cadence.com)

File Edit Tools Syntax Buffers Window Help

async_fifo_env.sv coverage.sv fifo_pkg.sv gray_bind_coverage.sv scoreboard.sv

```
	function void write_wr(fifo_transaction tx);
	      `uvm_info("SB_WRITE_CP","at this time it will capture a write packet",UVM_LOW)
	       if(tx.wr_en && !tx.wfull) begin
	               if(exp_q.size()>=FIFO_DEPTH) begin
	                       `uvm_error("sb","fifo OVERFLOW Attempted to write when FIFO is full")
	               end
	               else begin
	                       exp_q.push_back(tx.wdata);
	                       total_writes++;
	                       `uvm_info("scoreboard",$sformatf("write recieved write:data=%0h stored queue size=%0d",tx.wdata,exp_q.size()),UVM_MEDIUM)
	                       `uvm_info("sb",$sformatf("total count %0d",total_writes),UVM_LOW)
	               end
	        end
	endfunction

	function void write_rd(fifo_transaction tx);
	               if(tx.rd_en && !tx.rempty) begin
	                       if(exp_q.size()==0) begin
	                               `uvm_error("SCOREBOARD","queue empty but read happened")
	                               return;
	                       end
	                       req_q=exp_q.pop_front();
	                               `uvm_info("scoreboard",$sformatf("read recieved queue size=%0d",exp_q.size()),UVM_LOW)
	                       if(req_q==tx.rdata) begin
	                               match_count++;
	                               `uvm_info("fifo_sb",$sformatf("data matched  expected %0h,got %0h ,match count=%0d",req_q,tx.rdata,match_count),UVM_LOW)
	                       end
	                       else begin
	                               mismatch_count++;
	                               `uvm_error("fifo_sb",$sformatf("data mismatched expected %0h ,got %0h,mismatch count=%0d ",req_q,tx.rdata,mismatch_count))
	                       end

	               end
	endfunction

	function void reset();
	        exp_q.delete();
	        total_writes=0;
	        total_reads=0;
```

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scoreboard.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/env) (5 of 5) - GVIM (on ip-10-115-24-57.cadence.com)

File Edit Tools Syntax Buffers Window Help

async_fifo_env.sv coverage.sv fifo_pkg.sv gray_bind_coverage.sv scoreboard.sv

```
                        else begin
                                mismatch_count++;
                                `uvm_error("fifo_sb",$sformatf("data mismatched expected %0h ,got %0h,mismatch count=%0d ",req_q,tx.rdata,mismatch_count))
                        end

                end
        endfunction

        function void reset();
                exp_q.delete();
                total_writes=0;
                total_reads=0;
                mismatch_count=0;
                match_count=0;
                total_valid_reads=0;
                total_compares=0;
        endfunction

        function void report_phase(uvm_phase phase);
               `uvm_info("scoreboard report",$sformatf("final report:matched %0d ,mismatched:%0d",match_count,mismatch_count),UVM_NONE)
               if(mismatch_count==0)
                        `uvm_info("RESULT","TEST PASSES",UVM_NONE)
               else
                        `uvm_error("RESULT","TEST FAILED")
        endfunction

endclass
```

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/sequences) - GVIM6 (on ip-10-115-24-57.cadence

File Edit Tools Syntax Buffers Window Help

```
class wr_smoke_seq extends uvm_sequence #(fifo_transaction);

        `uvm_object_utils(wr_smoke_seq)

        int unsigned N=16;
        int write_count;
        function new(string name="wr_smoke_seq");
                super.new(name);
        endfunction

        virtual task body();
                fifo_transaction tx;
                write_count=0;

                `uvm_info(get_type_name(),$sformatf("starting smoke write-%0d words",N),UVM_MEDIUM)

                repeat(N) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with {
                                rd_en==0;
                                wr_en==1'b1;
                                wdata inside {[32'h0:32'hFFFF_FFFE]};
                                clr_overflow==1'b0;});
                        finish_item(tx);
                                #50;
                        write_count++;
                        `uvm_info(get_type_name(),$sformatf("write data=0x%08h data count=%0d",tx.wdata,write_count),UVM_HIGH)
                        #20;
                end
                repeat(1) begin
                        `uvm_do_with(tx,{wr_en==1'b0;rd_en==1'b0;wdata==0;clr_overflow==1;clr_underflow==1;})
                end
                        `uvm_info(get_type_name(),"smoke write completed",UVM_MEDIUM)
                        `uvm_info("seq","sequence started",UVM_LOW)
                endtask
endclass

class rd_smoke_seq extends uvm_sequence #(fifo_transaction);
        `uvm_object_utils(rd_smoke_seq)

        int unsigned N=16;
        int read_count;

        function new(string name="rd_smoke_seq");
                super.new(name);
        endfunction

        virtual task body();
                fifo_transaction tx;
                read_count=0;

                `uvm_info(get_type_name(),$sformatf("starting smoke read-%0d words",N),UVM_MEDIUM)
```

Applications Places System

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/sequences) - GVIM6 (on ip-10-115-24-57.cadence

File Edit Tools Syntax Buffers Window Help

```
	`uvm_object_utils(rd_smoke_seq)

	int unsigned N=16;
	int read_count;

	function new(string name="rd_smoke_seq");
		super.new(name);
	endfunction

	virtual task body();
		fifo_transaction tx;
		read_count=0;

		`uvm_info(get_type_name(),$sformatf("starting smoke read-%0d words",N),UVM_MEDIUM)

		repeat(N) begin
			tx=fifo_transaction::type_id::create("tx");
			start_item(tx);
			assert(tx.randomize() with {
				rd_en==1'b1;
				wr_en==0;
				clr_underflow==0;});
			finish_item(tx);
	  #50;
			read_count++;
			`uvm_info("seq",$sformatf("rd_en=%0d",tx.rd_en),UVM_LOW)
			`uvm_info(get_type_name(),$sformatf("read  rd_data=0x%08h rempty=%0d data count=%0d",tx.rdata,tx.rempty,read_count),UVM_HIGH)
		end
		repeat(2) begin
			tx=fifo_transaction::type_id::create("tx");
			start_item(tx);
			assert(tx.randomize() with {
				rd_en==1'b1;
				wr_en==0;
				clr_overflow==0;
				clr_underflow==0;});
			finish_item(tx);
	  #50;
			read_count++;
			`uvm_info("seq",$sformatf("rd_en=%0d",tx.rd_en),UVM_LOW)
			`uvm_info(get_type_name(),$sformatf("read  rd_data=0x%08h rempty=%0d data count=%0d",tx.rdata,tx.rempty,read_count),UVM_HIGH)
		end

		repeat(1) begin
			`uvm_do_with(tx,{wr_en==0;rd_en==0;clr_overflow==1;clr_underflow==1;})
		end
			`uvm_info(get_type_name(),"smoke read completed",UVM_MEDIUM)
			`uvm_info("seq","sequence started",UVM_LOW)

		endtask
endclass
```

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/sequences) - GVIM6 (on

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```
class random_wseq extends uvm_sequence #(fifo_transaction);
        `uvm_object_utils(random_wseq)

        int unsigned num_transactions=1000;
        int write_count;

        function new(string name="random_wseq");
                super.new(name);
        endfunction

        virtual task body();
                fifo_transaction tx;
                write_count=0;
                repeat(num_transactions) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {
                                wr_en==1;
                                rd_en==0;
                                wdata inside {[32'h0:32'h7FFF_FFFF]};

                                clr_underflow==0;
                                clr_overflow==0;
                        });
                        finish_item(tx);
                        write_count++;
                end
                repeat(1) begin
                        `uvm_do_with(tx,{wr_en==0;rd_en==0;wdata==32'h0;clr_overflow==1;clr_underflow==1;})
                end
        endtask
endclass

class random_rseq extends uvm_sequence #(fifo_transaction);
        `uvm_object_utils(random_rseq)

        int unsigned num_transactions=1000;
        int read_count;

        function new(string name="random_rseq");
                super.new(name);
        endfunction

        virtual task body();
                fifo_transaction tx;
                read_count=0;
                repeat(num_transactions) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {
                                wr_en==0;
```

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/Asyn

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```
        endfunction

        virtual task body();
                fifo_transaction tx;
                read_count=0;
                repeat(num_transactions) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {
                                wr_en==0;
                                rd_en==1;
                                clr_overflow==0;
                                clr_underflow==0;
                        });
                        finish_item(tx);
                        read_count++;

                end
                repeat(3) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==0;
                                });
                        finish_item(tx);
                end

                repeat(1) begin
                        `uvm_do_with(tx,{wr_en==0;rd_en==0;clr_overflow==1;clr_underflow==1;})
                end

        endtask
endclass


class burst_write_seq extends uvm_sequence #(fifo_transaction);
        `uvm_object_utils(burst_write_seq)

        int write_count;
        int read_count;

        function new(string name="burst_write_seq");
                super.new(name);
        endfunction

        virtual task body();
                fifo_transaction tx;
                write_count=0;
                read_count=0;
                `uvm_info("seq","burst fill starts",UVM_MEDIUM)
```

File Edit Tools Syntax Buffers Window Help

```
class burst_write_seq extends uvm_sequence #(fifo_transaction);
        `uvm_object_utils(burst_write_seq)

        int write_count;
        int read_count;

        function new(string name="burst_write_seq");
                super.new(name);
        endfunction

        virtual task body();
                fifo_transaction tx;
                write_count=0;
                read_count=0;
                `uvm_info("seq","burst fill starts",UVM_MEDIUM)
                repeat(50) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==1'b1;
                                 rd_en==0;
                                 wdata inside {[32'h0:32'hFFFF_FFFF]};
                                 clr_underflow==0;
                                 clr_overflow==0;
                                });
                        finish_item(tx);
                end
                write_count++;
                repeat(5) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with{
                                 wr_en==0;
                                 rd_en==0;
                                });
                        finish_item(tx);
                end

                `uvm_info("seq","slow drain starts",UVM_MEDIUM)
                repeat(60) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==1'b1;
                        });
                        finish_item(tx);
                end
                read_count++;
                repeat(3) begin
                        tx=fifo_transaction::type_id::create("tx");
```

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/sequenc

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```
                `uvm_info("seq","slow drain starts",UVM_MEDIUM)
                repeat(60) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==1'b1;
                        });
                        finish_item(tx);
                end
                read_count++;
                repeat(3) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==0;
                               });
                        finish_item(tx);
                end


                repeat(1) begin
                        `uvm_do_with(tx,{wr_en==0;rd_en==0;clr_overflow==1;clr_underflow==1;})
                end


                `uvm_info("seq","sequence completed",UVM_MEDIUM)


        endtask
endclass
class burst_read_seq extends uvm_sequence #(fifo_transaction);
        `uvm_object_utils(burst_read_seq)

        int write_count;
        int read_count;

        function new(string name="burst_read_seq");
                super.new(name);
        endfunction

        task body();
                fifo_transaction tx;
                write_count=0;
                read_count=0;
                `uvm_info("seq","slow fill starts",UVM_MEDIUM)
                repeat(70) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==1'b1;
```

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verificatio

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```
class burst_read_seq extends uvm_sequence #(fifo_transaction);
        `uvm_object_utils(burst_read_seq)

        int write_count;
        int read_count;

        function new(string name="burst_read_seq");
                super.new(name);
        endfunction

        task body();
                fifo_transaction tx;
                write_count=0;
                read_count=0;
                `uvm_info("seq","slow fill starts",UVM_MEDIUM)
                repeat(70) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==1'b1;
                                 rd_en==0;
                                 wdata inside {[32'h0:32'hFFFF_FFFF]};
                                 clr_underflow==0;
                                 clr_overflow==0;
                                });
                        finish_item(tx);
                end
                write_count++;
                repeat(3) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==0;
                                });
                        finish_item(tx);
                end
                `uvm_info("seq","burst drain starts",UVM_MEDIUM)
                repeat(65) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==1'b1;
                                 clr_underflow==0;
                                 clr_overflow==0;
                                });
                        finish_item(tx);
                end
                read_count++;
                repeat(15) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
```

Applications Places System

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/sequences)

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```
                end
                read_count++;
                repeat(15) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==1;
                                });
                        finish_item(tx);
                end
                repeat(3) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==0;
                                });
                        finish_item(tx);
                end
                repeat(1) begin
                        `uvm_do_with(tx,{wr_en==0;rd_en==0;clr_overflow==1;clr_underflow==1;wdata==32'h0;rdata==32'h0;})
                end



        endtask
endclass
class simultaneous_wseq extends uvm_sequence #(fifo_transaction);
        `uvm_object_utils(simultaneous_wseq)

        int write_count;
        int read_count;

        function new(string name="simultaneous_wseq");
                super.new(name);
        endfunction

        task body();
                fifo_transaction tx;
                write_count=0;
                repeat(40) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==1'b1;
                                 rd_en==0;
                                 wdata inside {32'h0000_0000,32'hFFFF_FFFF};
                                 clr_underflow==0;
                                 clr_overflow==0;
                                });
                        finish_item(tx);
                        write_count++;
```

Applications Places System

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/Asy

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```
endfunction

task body();
        fifo_transaction tx;
        write_count=0;
        repeat(40) begin
                tx=fifo_transaction::type_id::create("tx");
                start_item(tx);
                assert(tx.randomize() with
                        {wr_en==1'b1;
                         rd_en==0;
                         wdata inside {32'h0000_0000,32'hFFFF_FFFF};
                         clr_underflow==0;
                         clr_overflow==0;
                        });
                finish_item(tx);
                write_count++;

        end
        repeat(20) begin
                tx=fifo_transaction::type_id::create("tx");
                start_item(tx);
                assert(tx.randomize() with
                        {wr_en==1'b1;
                         rd_en==0;
                         wdata=={32'h5555_5555};
                         clr_underflow==0;
                         clr_overflow==0;
                        });
                finish_item(tx);
                write_count++;

        end
        repeat(30) begin
                tx=fifo_transaction::type_id::create("tx");
                start_item(tx);
                assert(tx.randomize() with
                        {wr_en==1'b1;
                         rd_en==0;
                         wdata inside {[32'h0001_0000:32'hFFFF_FFFF]};
                         clr_underflow==0;
                         clr_overflow==0;
                        });
                finish_item(tx);
                write_count++;

        end
        repeat(30) begin
                tx=fifo_transaction::type_id::create("tx");
                start_item(tx);
                assert(tx.randomize() with
                        {wr_en==1'b1;
                         rd_en==0;
```

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```
                end
                repeat(30) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==1'b1;
                                 rd_en==0;
                                 wdata=={32'hAAAA_AAAA};
                                 clr_underflow==0;
                                 clr_overflow==0;
                                });
                        finish_item(tx);
                        write_count++;

                end
                repeat(3) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==0;
                                });
                        finish_item(tx);
                end

                repeat(1) begin
                        `uvm_do_with(tx,{wr_en==1'b0;rd_en==1'b0;wdata==0;clr_overflow==1;clr_underflow==1;})
                end

        endtask
endclass

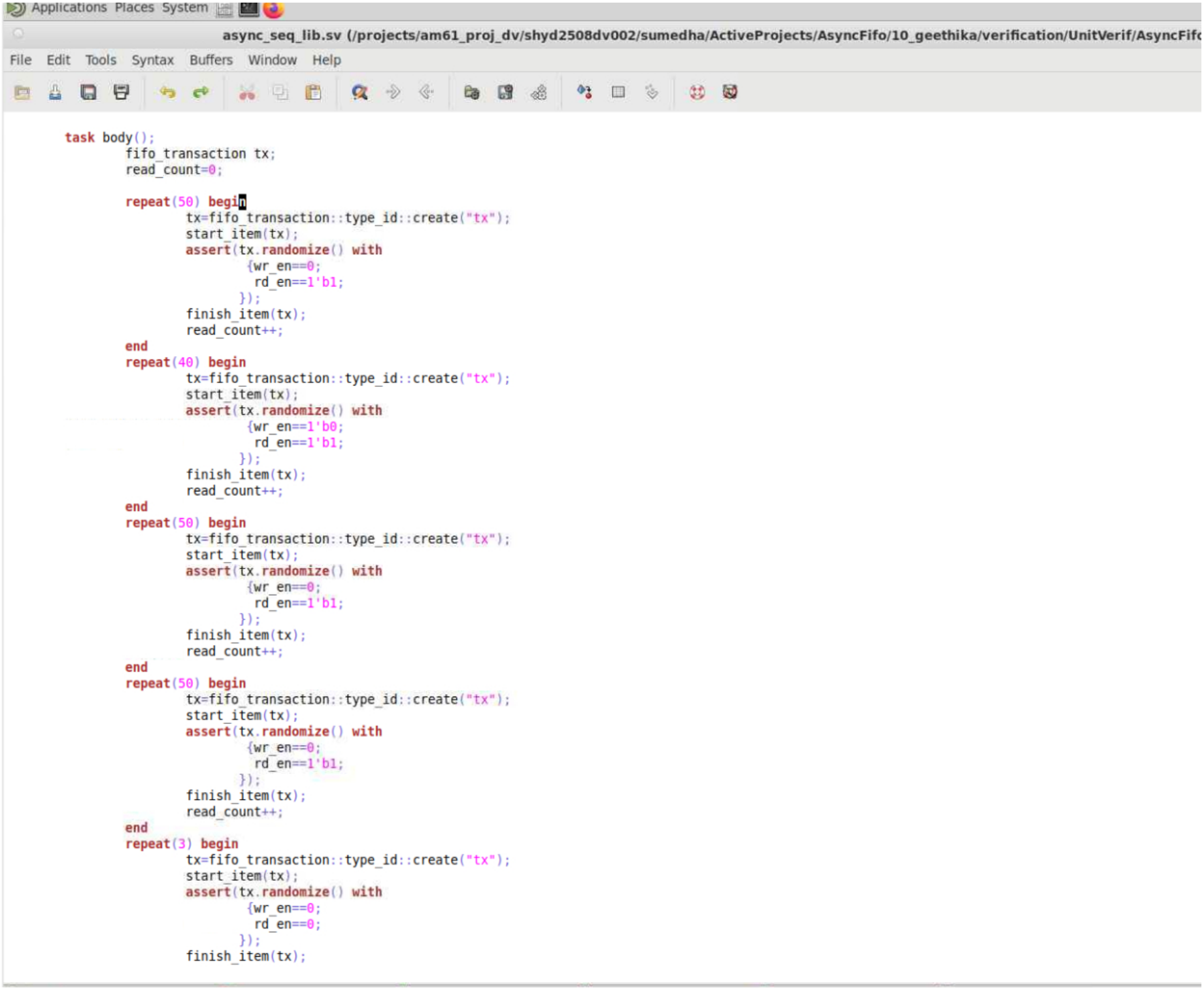
class simultaneous_rseq extends uvm_sequence #(fifo_transaction);
        `uvm_object_utils(simultaneous_rseq)

        int read_count;


        function new(string name="simultaneous_rseq");
                super.new(name);
        endfunction

        task body();
                fifo_transaction tx;
                read_count=0;

                repeat(50) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==1'b1;
```


Applications Places System
async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo
File Edit Tools Syntax Buffers Window Help
task body();
fifo_transaction tx;
read_count=0;
repeat(50) begin
tx=fifo_transaction::type_id::create("tx");
start_item(tx);
assert(tx.randomize() with
{wr_en==0;
rd_en==1'b1;
});
finish_item(tx);
read_count++;
end
repeat(40) begin
tx=fifo_transaction::type_id::create("tx");
start_item(tx);
assert(tx.randomize() with
{wr_en==1'b0;
rd_en==1'b1;
});
finish_item(tx);
read_count++;
end
repeat(50) begin
tx=fifo_transaction::type_id::create("tx");
start_item(tx);
assert(tx.randomize() with
{wr_en==0;
rd_en==1'b1;
});
finish_item(tx);
read_count++;
end
repeat(50) begin
tx=fifo_transaction::type_id::create("tx");
start_item(tx);
assert(tx.randomize() with
{wr_en==0;
rd_en==1'b1;
});
finish_item(tx);
read_count++;
end
repeat(3) begin
tx=fifo_transaction::type_id::create("tx");
start_item(tx);
assert(tx.randomize() with
{wr_en==0;
rd_en==0;
});
finish_item(tx);

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncF

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```
                        read_count++;
                end
                repeat(50) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==1'b1;
                                });
                        finish_item(tx);
                        read_count++;
                end
                repeat(3) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==0;
                                });
                        finish_item(tx);
                end

                repeat(1) begin
                        `uvm_do_with(tx,{wr_en==1'b0;rd_en==1'b0;wdata==0;clr_overflow==1;clr_underflow==1;})
                end
        endtask
endclass

class threshold_seq extends uvm_sequence #(fifo_transaction);
        `uvm_object_utils(threshold_seq)


        function new(string name="threshold_seq");
                super.new(name);
        endfunction

        task body();
                fifo_transaction tx;
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==1;
                                 rd_en==0;
                                wdata inside {[32'h0001_0000:32'h7FFF_FFFE]};
                                });
                        finish_item(tx);
                repeat(30) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {
                                wr_en dist{1:=70,0:=30};
                                rd_en==1;
```

Applications Places System

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/Unit

File Edit Tools Syntax Buffers Window Help

```
        finish_item(tx);
    repeat(30) begin
        tx=fifo_transaction::type_id::create("tx");
        start_item(tx);
        assert(tx.randomize() with
                {
                wr_en dist{1:=70,0:=30};
                rd_en==1;
                wdata inside {[32'h0:32'h5555_5555]};
                });
        finish_item(tx);
    end
    repeat(30) begin
        tx=fifo_transaction::type_id::create("tx");
        start_item(tx);
        assert(tx.randomize() with
                {
                wr_en dist{1:=70,0:=30};
                rd_en==0;
                wdata inside {[32'h5555_5555:32'hAAAA_AAAA]};
                });
        finish_item(tx);
    end

    repeat(30) begin
        tx=fifo_transaction::type_id::create("tx");
        start_item(tx);
        assert(tx.randomize() with
                {
                wr_en dist{1:=70,0:=30};
                rd_en==1;
                wdata inside {[32'hAAAA_AAAA:32'hFFFF_FFFF]};
                });
        finish_item(tx);
    end

    repeat(150) begin
        tx=fifo_transaction::type_id::create("tx");
        start_item(tx);
        assert(tx.randomize() with
                {
                wr_en==0;
                rd_en dist{1:=50,0:=50};
                });
        finish_item(tx);
    end
    repeat(3) begin
        tx=fifo_transaction::type_id::create("tx");
        start_item(tx);
        assert(tx.randomize() with
                {wr_en==0;
                 rd_en==0;
                });
```

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/sequences) -

File Edit Tools Syntax Buffers Window Help

```
                        rd_en dist{1:=50,0:=50};
                        });
                finish_item(tx);
        end
        repeat(3) begin
                tx=fifo_transaction::type_id::create("tx");
                start_item(tx);
                assert(tx.randomize() with
                        {wr_en==0;
                         rd_en==0;
                        });
                finish_item(tx);
        end

        repeat(1) begin
                `uvm_do_with(tx,{wr_en==1'b0;rd_en==1'b0;wdata==0;clr_overflow==1;clr_underflow==1;})
        end
    endtask
endclass


class reset_seq extends uvm_sequence #(fifo_transaction);
        `uvm_object_utils(reset_seq)

        virtual fifo_if vif;
        function new(string name="reset_seq");
                super.new(name);
        endfunction

        task body();
                fifo_transaction tx;
                if(!uvm_config_db#(virtual fifo_if)::get(null,"","vif",vif))
                        `uvm_fatal("seq","no vif")
                repeat(20) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with{
                                wrst_n==0;
                                rrst_n==0;
                                wr_en ==1;
                                rd_en ==1;
                        });
                        finish_item(tx);
                        #5;
                end
                repeat(20) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with{
                                wrst_n==1;
                                rrst_n==0;
                                wr_en==0;
                                rd_en ==1;
```

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/sequences) - GVIM6

File Edit Tools Syntax Buffers Window Help

```
                repeat(20) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with{
                                wrst_n==1;
                                rrst_n==0;
                                wr_en==0;
                                rd_en ==1;
                        });
                        finish_item(tx);
                        #5;
                end
                repeat(20) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with{
                                rrst_n==1;
                                wrst_n==0;
                                wr_en ==0;
                                rd_en ==1;
                        });
                        finish_item(tx);
                        #5;
                end
                repeat(46) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with{
                                rrst_n==1;
                                wrst_n==1;
                                wr_en ==1;
                                rd_en ==1;
                        });
                finish_item(tx);
                        #5;
                end
                repeat(3) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==0;
                                });
                        finish_item(tx);
                end

                `uvm_info("seq","applying  write reset ",UVM_MEDIUM)
                repeat(1) begin
                        `uvm_do_with(tx,{wr_en==1'b0;rd_en==1'b0;wdata==0;clr_overflow==1;clr_underflow==1;})
                end
        endtask
endclass
```

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/sequences) - GVIM6 (on ip-10-115-24-

File Edit Tools Syntax Buffers Window Help

```
endclass

class overflow_underflow_seq extends uvm_sequence #(fifo_transaction);
        `uvm_object_utils(overflow_underflow_seq)


        function new(string name="overflow_underflow_seq");
                super.new(name);
        endfunction

        task body();
                fifo_transaction tx;
                repeat(300) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {
                                wr_en==1;
                                rd_en==0;
                                wdata inside {[32'h0:32'h7FFF_FFFE]};
                               });
                        finish_item(tx);
                end
                repeat(330) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {
                                rd_en==1'b1;
                                wr_en==0;
                               });
                        finish_item(tx);
                end
                repeat(3) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==0;
                               });
                        finish_item(tx);
                end

                repeat(1) begin
                        `uvm_do_with(tx,{wr_en==1'b0;rd_en==1'b0;wdata==0;clr_overflow==1;clr_underflow==1;})
                end

        endtask
endclass
```

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/s

File Edit Tools Syntax Buffers Window Help

```
class phase_sweep_seq extends uvm_sequence #(fifo_transaction);
        `uvm_object_utils(phase_sweep_seq)
        virtual fifo_if vif;

        function new(string name="overflow_underflow_seq");
                super.new(name);
        endfunction

        task body();
                fifo_transaction tx;
                int phase;
                phase=0;
                repeat(70) begin
                        `uvm_info("seq",$sformatf("applying phase offset =%0d",phase),UVM_LOW)

                repeat(90) begin
                                tx=fifo_transaction::type_id::create("tx");
                                start_item(tx);
                                assert(tx.randomize() with
                                        {
                                        wr_en dist{1:=70,0:=30};
                                        rd_en==0;
                                        });
                                        finish_item(tx);
                        end
                phase=phase+1;
        end
        repeat(70) begin
                        `uvm_info("seq",$sformatf("applying phase offset =%0d",phase),UVM_LOW)


                        repeat(90) begin
                                tx=fifo_transaction::type_id::create("tx");
                                start_item(tx);
                                assert(tx.randomize() with
                                        {
                                        wr_en==0;
                                        rd_en==!wr_en;
                                        });
                                        finish_item(tx);
                        end
                phase=phase+1;
        end
        repeat(3) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==0;
                                });
                        finish_item(tx);
                end
```

File Edit Tools Syntax Buffers Window Help

```
class fifo_full_overflow_seq extends uvm_sequence #(fifo_transaction);
        `uvm_object_utils(fifo_full_overflow_seq)

        function new(string name="fifo_full_overflow_seq");
                super.new(name);
        endfunction

        task body();
                fifo_transaction tx;
                repeat(30) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==1'b1;
                                 rd_en==0;
                                 wdata inside {[32'h0:32'hFFFF_FFFF]};
                                });
                        finish_item(tx);
        end

                repeat(8) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==1;
                                 rd_en==0;
                                });
                        finish_item(tx);
                end
                repeat(3) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==0;
                                });
                        finish_item(tx);
                end

                repeat(1) begin
                        `uvm_do_with(tx,{wr_en==1'b0;rd_en==1'b0;wdata==0;clr_overflow==1;clr_underflow==1;})
                end
    endtask
endclass
```

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/sequences) - GVIM6 (on ip-10-115-24-57.cader

File Edit Tools Syntax Buffers Window Help

```
class fifo_empty_underflow_seq extends uvm_sequence #(fifo_transaction);
        `uvm_object_utils(fifo_empty_underflow_seq)


        function new(string name="fifo_empty_underflow_seq");
                super.new(name);
        endfunction

        task body();
                fifo_transaction tx;

                repeat(20) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==1'b1;
                                 rd_en==0;
                                 wdata inside {[32'h0:32'hFFFF_FFFF]};
                                });
                        finish_item(tx);
                end

                repeat(20/2) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==1;
                                 rd_en==0;
                                });
                        finish_item(tx);
                end
                        repeat(20/2) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==1;
                                 rd_en==0;
                                 wdata inside {[32'h0:32'hFFFF_FFFF]};
                                });
                        finish_item(tx);
                end
                repeat(3) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==0;
                                });
                        finish_item(tx);
                end

                repeat(1) begin
                        `uvm_do_with(tx,{wr_en==1'b0;rd_en==1'b0;wdata==0;clr_overflow==1;clr_underflow==1;})
```

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/sequences) - GV

File Edit Tools Syntax Buffers Window Help

```
                repeat(1) begin
                        `uvm_do_with(tx,{wr_en==1'b0;rd_en==1'b0;wdata==0;clr_overflow==1;clr_underflow==1;})
                end

    endtask
endclass

class fifo_empty_seq extends uvm_sequence #(fifo_transaction);
        `uvm_object_utils(fifo_empty_seq)

        function new(string name="fifo_empty_seq");
                super.new(name);
        endfunction

        task body();
                fifo_transaction tx;

                repeat(30) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==1'b1;
                                 rd_en==0;
                                });
                        finish_item(tx);
                end

                repeat(40) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==1'b1;
                                });
                        finish_item(tx);
                end
                repeat(3) begin
                        tx=fifo_transaction::type_id::create("tx");
                        start_item(tx);
                        assert(tx.randomize() with
                                {wr_en==0;
                                 rd_en==0;
                                });
                        finish_item(tx);
                end
                repeat(1) begin
                        `uvm_do_with(tx,{wr_en==1'b0;rd_en==1'b0;wdata==0;clr_overflow==1;clr_underflow==1;})
                end

    endtask
endclass
```

async_seq_lib.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/sequences) - GV

File Edit Tools Syntax Buffers Window Help

```
    endtask
endclass

class fifo_full_seq extends uvm_sequence #(fifo_transaction);
       `uvm_object_utils(fifo_full_seq)

       function new(string name="fifo_empty_seq");
              super.new(name);
       endfunction

       task body();
              fifo_transaction tx;

              for(int i=0;i<16;i++) begin
                     tx=fifo_transaction::type_id::create("tx");
                     start_item(tx);
                     assert(tx.randomize() with
                            {wr_en==1'b1;
                             rd_en==0;
                             wdata inside {[32'h0000_0000:32'h7FFF_FFFE]};
                            });
                     finish_item(tx);
              end

              for(int i=0;i<16;i++) begin
                     tx=fifo_transaction::type_id::create("tx");
                     start_item(tx);
                     assert(tx.randomize() with
                            {wr_en==0;
                             rd_en==1'b1;
                            });
                     finish_item(tx);
              end
              repeat(3) begin
                     tx=fifo_transaction::type_id::create("tx");
                     start_item(tx);
                     assert(tx.randomize() with
                            {wr_en==0;
                             rd_en==0;
                            });
                     finish_item(tx);
              end
              repeat(1) begin
                     `uvm_do_with(tx,{wr_en==1'b0;rd_en==1'b0;wdata==0;clr_overflow==1;clr_underflow==1;})
              end

    endtask
endclass

~
~
```

asyn_fifo_tests.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/tests) -

File Edit Tools Syntax Buffers Window Help

```
class async_base_test extends uvm_test;
        async_fifo_env env;
        virtual fifo_if vif;
        `uvm_component_utils(async_base_test)

        function new(string name="async_base_test",uvm_component parent);
                super.new(name,parent);
        endfunction

        virtual function void build_phase(uvm_phase phase);
                super.build_phase(phase);

                if(!uvm_config_db #(virtual fifo_if)::get(this,"","vif",vif)) begin
                        `uvm_fatal("test","virtual interface not set")
                end
                env=async_fifo_env::type_id::create("env",this);

        endfunction
        virtual function void end_of_elaboration_phase(uvm_phase phase);
                super.end_of_elaboration_phase(phase);

                uvm_top.print_topology();
        endfunction

        task reset_phase(uvm_phase phase);
                phase.raise_objection(this);
                `uvm_info(get_type_name(),"applying write reset",UVM_MEDIUM)
                apply_wreset();
                `uvm_info(get_type_name(),"applying read reset",UVM_MEDIUM)
                apply_rreset();
                `uvm_info(get_type_name(),"reset complete",UVM_MEDIUM)
                phase.drop_objection(this);
        endtask

        virtual task apply_wreset();
                virtual fifo_if vif;
                if(!uvm_config_db#(virtual fifo_if)::get(this,"","vif",vif))
                        `uvm_fatal("NOVIF","virtual interface not found in base test")

                vif.wrst_n<=0;
                repeat(5) @(posedge vif.wclk);
                vif.wrst_n<=1;
                repeat(2) @(posedge vif.wclk);
        endtask

        virtual task apply_rreset();
                virtual fifo_if vif;
```

-- INSERT --

asyn_fifo_tests.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/tests) -

File Edit Tools Syntax Buffers Window Help

```
class smoke_test extends async_base_test;


        `uvm_component_utils(smoke_test)

        function new(string name,uvm_component parent);
                super.new(name,parent);
        endfunction


        virtual task run_phase(uvm_phase phase);

        wr_smoke_seq wr_seq;
        rd_smoke_seq rd_seq;

        phase.raise_objection(this);

                `uvm_info(get_type_name(),"running reset value verification test",UVM_MEDIUM)
                `uvm_info("test","run phase strted",UVM_LOW)

                fork
                        apply_wreset();
                        apply_rreset();
                join
                wr_seq=wr_smoke_seq::type_id::create("wr_seq");
                rd_seq=rd_smoke_seq::type_id::create("rd_seq");
                env.cg.sample_clk_ratio(vif.wclk_period,vif.rclk_period);

                wr_seq.start(env.wr_ag.wr_sqr);
                #50;
                rd_seq.start(env.rd_ag.rd_sqr);
                #50;
                `uvm_info(get_type_name(),"smoke test completed",UVM_NONE)
                #50;
        phase.drop_objection(this);

        endtask
endclass
```

asyn_fifo_tests.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/tests) - GVIM4 (on ip-10-115-2

File Edit Tools Syntax Buffers Window Help

```
        endtask
endclass

class random_test extends async_base_test;


        `uvm_component_utils(random_test)

        function new(string name,uvm_component parent);
                super.new(name,parent);
        endfunction


        virtual task run_phase(uvm_phase phase);

        random_wseq wseq;
        random_rseq rseq;

        phase.raise_objection(this);

                `uvm_info(get_type_name(),"running reset value verification test",UVM_MEDIUM)
                `uvm_info("test","run phase strted",UVM_LOW)

                fork
                        apply_wreset();
                        apply_rreset();
                join

                wseq=random_wseq::type_id::create("wseq");
                rseq=random_rseq::type_id::create("rseq");
                env.cg.sample_clk_ratio(vif.wclk_period,vif.rclk_period);

                wseq.start(env.wr_ag.wr_sqr);
                #50;
                rseq.start(env.rd_ag.rd_sqr);

                #50;
                `uvm_info(get_type_name(),"random test completed",UVM_NONE)
        phase.drop_objection(this);

        endtask
endclass
```

asyn_fifo_tests.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/tests) - GVIM4 (on ip

File Edit Tools Syntax Buffers Window Help

```
        endtask
endclass

class burst_write_test extends async_base_test;


        `uvm_component_utils(burst_write_test)

        function new(string name,uvm_component parent);
                super.new(name,parent);
        endfunction


        virtual task run_phase(uvm_phase phase);

        burst_write_seq wseq;

        phase.raise_objection(this);

                `uvm_info(get_type_name(),"running reset value verification test",UVM_MEDIUM)
                `uvm_info("test","run phase strted",UVM_LOW)

                fork
                        apply_wreset();
                        apply_rreset();
                join

                wseq=burst_write_seq::type_id::create("wseq");
                env.cg.sample_clk_ratio(vif.wclk_period,vif.rclk_period);

                wseq.start(env.wr_ag.wr_sqr);
                #50;
                wseq.start(env.rd_ag.rd_sqr);
                #50;
                `uvm_info(get_type_name(),"burst write to full test completed",UVM_NONE)
        phase.drop_objection(this);

        endtask
endclass
```

asyn_fifo_tests.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/tests) - GVIM4 (on ip-10-11

File Edit Tools Syntax Buffers Window Help

```
class burst_read_test extends async_base_test;

        `uvm_component_utils(burst_read_test)

        function new(string name,uvm_component parent);
                super.new(name,parent);
        endfunction


        virtual task run_phase(uvm_phase phase);

        burst_read_seq rseq;

        phase.raise_objection(this);

                `uvm_info(get_type_name(),"running reset value verification test",UVM_MEDIUM)
                `uvm_info("test","run phase strted",UVM_LOW)

                fork
                        apply_wreset();
                        apply_rreset();
                join

                rseq=burst_read_seq::type_id::create("rseq");
                env.cg.sample_clk_ratio(vif.wclk_period,vif.rclk_period);

                rseq.start(env.wr_ag.wr_sqr);
                #50;
                rseq.start(env.rd_ag.rd_sqr);
                #50;
                `uvm_info(get_type_name(),"burst read to empty test completed",UVM_NONE)
        phase.drop_objection(this);

        endtask
endclass

class simultaneous_test extends async_base_test;


        `uvm_component_utils(simultaneous_test)

        function new(string name,uvm_component parent);
                super.new(name,parent);
        endfunction
```

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File Edit

```
class simultaneous_test extends async_base_test;


        `uvm_component_utils(simultaneous_test)

        function new(string name,uvm_component parent);
                super.new(name,parent);
        endfunction


        virtual task run_phase(uvm_phase phase);

        simultaneous_wseq wseq;
        simultaneous_rseq rseq;

        phase.raise_objection(this);

                `uvm_info(get_type_name(),"running reset value verification test",UVM_MEDIUM)
                `uvm_info("test","run phase strted",UVM_LOW)

                fork
                        apply_wreset();
                        apply_rreset();
                join

                wseq=simultaneous_wseq::type_id::create("wseq");
                rseq=simultaneous_rseq::type_id::create("rseq");
                env.cg.sample_clk_ratio(vif.wclk_period,vif.rclk_period);

                fork
                wseq.start(env.wr_ag.wr_sqr);
                rseq.start(env.rd_ag.rd_sqr);
                join

                #10;
                `uvm_info(get_type_name(),"random test completed",UVM_NONE)
        phase.drop_objection(this);

        endtask
endclass

class clk_ratio_test extends async_base_test;


        `uvm_component_utils(clk_ratio_test)

        simultaneous_wseq wseq;
-- INSERT --
```

asyn_fifo_tests.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/tests) - GVIM4 (on ip-10-115-24-57.cadence.com)

File Edit Tools Syntax Buffers Window Help

```
class clk_ratio_test extends async_base_test;


        `uvm_component_utils(clk_ratio_test)

        simultaneous_wseq wseq;
        simultaneous_rseq rseq;


        function new(string name="clk_ratio_test",uvm_component parent);
                super.new(name,parent);
        endfunction


        task run_phase(uvm_phase phase);


        phase.raise_objection(this);

                `uvm_info(get_type_name(),"running reset value verification test",UVM_MEDIUM)
                `uvm_info("test","run phase strted",UVM_LOW)

                fork
                        apply_wreset();
                        apply_rreset();
                join

                for(int i=1;i<=10;i++) begin
                        int w_ratio=i;
                        int r_ratio=11-i;

                        vif.wclk_period=w_ratio*4;
                        vif.rclk_period=r_ratio*4;  //4 keeps simulation stable avoids too fast clks

                        `uvm_info("CLK_SLEEP",$sformatf("Running ratioW:R=%0d:%0d (period %0d:%0d)",w_ratio,r_ratio,vif.wclk_period,vif.rclk_period),UVM_LOW)

                        wseq=simultaneous_wseq::type_id::create("wseq");
                        rseq=simultaneous_rseq::type_id::create("rseq");
                        env.cg.sample_clk_ratio(vif.wclk_period,vif.rclk_period);

                        fork
                                begin
                                        wseq.start(env.wr_ag.wr_sqr);
                                end
                                begin
                                        rseq.start(env.rd_ag.rd_sqr);
                                end
```

-- INSERT --  287,7-21

asyn_fifo_tests.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/tests) - GVIM4 (on ip-10-115-24-57.cadenc

File Edit Tools Syntax Buffers Window Help

```
                        fork
                                begin
                                        wseq.start(env.wr_ag.wr_sqr);
                                end
                                begin
                                        rseq.start(env.rd_ag.rd_sqr);
                                end
                        join
                end
                `uvm_info(get_type_name(),"random test completed",UVM_NONE)
        phase.drop_objection(this);

        endtask
endclass

class threshold_test extends async_base_test;


        `uvm_component_utils(threshold_test)

        function new(string name,uvm_component parent);
                super.new(name,parent);
        endfunction


        task run_phase(uvm_phase phase);

        threshold_seq tseq;

        phase.raise_objection(this);

                `uvm_info(get_type_name(),"running reset value verification test",UVM_MEDIUM)
                `uvm_info("test","run phase strted",UVM_LOW)

                fork
                        apply_wreset();
                        apply_rreset();
                join

                tseq=threshold_seq::type_id::create("tseq");
                env.cg.sample_clk_ratio(vif.wclk_period,vif.rclk_period);

                tseq.start(env.wr_ag.wr_sqr);
                #50;
                tseq.start(env.rd_ag.rd_sqr);
                #50;
                `uvm_info(get_type_name(),"burst read to empty test completed",UVM_NONE)
-- INSERT --
```

asyn_fifo_tests.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/tests) - GVIM4 (on ip-10-11

File Edit Tools Syntax Buffers Window Help

```
                tseq.start(env.wr_ag.wr_sqr);
                #50;
                tseq.start(env.rd_ag.rd_sqr);
                #50;
                `uvm_info(get_type_name(),"burst read to empty test completed",UVM_NONE)
        phase.drop_objection(this);

        endtask
endclass


class reset_test extends async_base_test;


        `uvm_component_utils(reset_test)

        function new(string name,uvm_component parent);
                super.new(name,parent);
        endfunction


        virtual task run_phase(uvm_phase phase);

        reset_seq seq;

        phase.raise_objection(this);

                `uvm_info(get_type_name(),"running reset value verification test",UVM_MEDIUM)
                `uvm_info("test","run phase strted",UVM_LOW)

                fork
                        apply_wreset();
                        apply_rreset();
                join

                seq=reset_seq::type_id::create("seq");
                env.cg.sample_clk_ratio(vif.wclk_period,vif.rclk_period);

                seq.start(env.wr_ag.wr_sqr);
                #50;
                seq.start(env.rd_ag.rd_sqr);
                #50;
                `uvm_info(get_type_name(),"burst read to empty test completed",UVM_NONE)
        phase.drop_objection(this);

        endtask
endclass
```

asyn_fifo_tests.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/tests) - GVIM4 (on ip-10-1

File Edit Tools Syntax Buffers Window Help

```
endclass

class of_uf_test extends async_base_test;


        `uvm_component_utils(of_uf_test)


        function new(string name,uvm_component parent);
                super.new(name,parent);
        endfunction


        virtual task run_phase(uvm_phase phase);

        overflow_underflow_seq seq;

        phase.raise_objection(this);

                `uvm_info(get_type_name(),"running reset value verification test",UVM_MEDIUM)
                `uvm_info("test","run phase strted",UVM_LOW)

                fork
                        apply_wreset();
                        apply_rreset();
                join

                seq=overflow_underflow_seq::type_id::create("seq");
                env.cg.sample_clk_ratio(vif.wclk_period,vif.rclk_period);

                seq.start(env.wr_ag.wr_sqr);
                #50;
                seq.start(env.rd_ag.rd_sqr);
                #50;
                `uvm_info(get_type_name(),"burst read to empty test completed",UVM_NONE)
        phase.drop_objection(this);

        endtask
endclass
```

```
class phase_sweep_test extends async_base_test;


        `uvm_component_utils(phase_sweep_test)

        function new(string name,uvm_component parent);
                super.new(name,parent);
        endfunction


        virtual task run_phase(uvm_phase phase);

        phase_sweep_seq seq;

        phase.raise_objection(this);

                `uvm_info(get_type_name(),"running reset value verification test",UVM_MEDIUM)
                `uvm_info("test","run phase strted",UVM_LOW)

                fork
                        apply_wreset();
                        apply_rreset();
                join

                seq=phase_sweep_seq::type_id::create("seq");
                env.cg.sample_clk_ratio(vif.wclk_period,vif.rclk_period);

                seq.vif=env.vif;
                seq.start(env.wr_ag.wr_sqr);
                #50;
                seq.start(env.rd_ag.rd_sqr);
                #50;
                `uvm_info(get_type_name(),"burst read to empty test completed",UVM_NONE)
        phase.drop_objection(this);

        endtask
endclass


class fifo_flag_test extends async_base_test;


        `uvm_component_utils(fifo_flag_test)

        function new(string name,uvm_component parent);
                super.new(name,parent);
```

Applications Places System

asyn_fifo_tests.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/tests) - GVIM4 (on ip-10-115-24-57.cadence.com)

File Edit Tools Syntax Buffers Window Help

```
class fifo_flag_test extends async_base_test;

        `uvm_component_utils(fifo_flag_test)

        function new(string name,uvm_component parent);
                super.new(name,parent);
        endfunction


        virtual task run_phase(uvm_phase phase);

        fifo_full_overflow_seq fseq;
        fifo_empty_underflow_seq eseq;

        phase.raise_objection(this);

                `uvm_info(get_type_name(),"running reset value verification test",UVM_MEDIUM)
                `uvm_info("test","run phase strted",UVM_LOW)

                fork
                        apply_wreset();
                        apply_rreset();
                join

                fseq=fifo_full_overflow_seq::type_id::create("fseq");
                eseq=fifo_empty_underflow_seq::type_id::create("eseq");
                env.cg.sample_clk_ratio(vif.wclk_period,vif.rclk_period);

                fork
                begin

                fseq.start(env.wr_ag.wr_sqr);
                #50;
                fseq.start(env.rd_ag.rd_sqr);
                #50;
                end
                begin
                eseq.start(env.wr_ag.wr_sqr);
                #50;
                eseq.start(env.rd_ag.rd_sqr);
                #50;
                end
                join

                `uvm_info(get_type_name(),"full flag to overflow test completed",UVM_NONE)
        phase.drop_objection(this);
-- INSERT --
```

asyn_fifo_tests.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/tests) - GVIM4 (on ip-10-11

File Edit Tools Syntax Buffers Window Help

```
class fifo_empty_test extends async_base_test;


        `uvm_component_utils(fifo_empty_test)

        function new(string name,uvm_component parent);
                super.new(name,parent);
        endfunction


        virtual task run_phase(uvm_phase phase);

        fifo_empty_seq eseq;
        fifo_full_overflow_seq fseq;

        phase.raise_objection(this);

                `uvm_info(get_type_name(),"running reset value verification test",UVM_MEDIUM)
                `uvm_info("test","run phase strted",UVM_LOW)

                fork
                        apply_wreset();
                        apply_rreset();
                join

                fseq=fifo_full_overflow_seq::type_id::create("fseq");
                eseq=fifo_empty_seq::type_id::create("eseq");
                env.cg.sample_clk_ratio(vif.wclk_period,vif.rclk_period);

                fork
                begin

                fseq.start(env.wr_ag.wr_sqr);
                #50;
                fseq.start(env.rd_ag.rd_sqr);
                #50;
                end
                begin
                eseq.start(env.wr_ag.wr_sqr);
                #50;
                eseq.start(env.rd_ag.rd_sqr);
                #50;
                end
                join


                `uvm_info(get_type_name(),"burst read to empty test completed",UVM_NONE)
-- INSERT --
```

asyn_fifo_tests.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/tests) - GVIM4 (on ip-10-115-24-57.c

File Edit Tools Syntax Buffers Window Help

```
class fifo_full_test extends async_base_test;


        `uvm_component_utils(fifo_full_test)

        function new(string name,uvm_component parent);
                super.new(name,parent);
        endfunction


        virtual task run_phase(uvm_phase phase);

        fifo_full_seq fseq;
//      fifo_full_overflow_seq fseq;

        phase.raise_objection(this);

                `uvm_info(get_type_name(),"running reset value verification test",UVM_MEDIUM)
                `uvm_info("test","run phase strted",UVM_LOW)

                fork
                        apply_wreset();
                        apply_rreset();
                join

                fseq=fifo_full_seq::type_id::create("fseq");
        //      eseq=fifo_empty_seq::type_id::create("eseq");
                env.cg.sample_clk_ratio(vif.wclk_period,vif.rclk_period);

                fork
                begin

                fseq.start(env.wr_ag.wr_sqr);
                #50;
                fseq.start(env.rd_ag.rd_sqr);
                #50;
                end
                join


                `uvm_info(get_type_name(),"burst read to empty test completed",UVM_NONE)
        phase.drop_objection(this);

        endtask
endclass
```

async_fifo_env.sv | coverage.sv | fifo_pkg.sv | gray_bind_coverage.sv | scoreboard.sv

```
`uvm_analysis_imp_decl(_wrt)
`uvm_analysis_imp_decl(_rdd)

class coverage extends uvm_component;
        `uvm_component_utils(coverage)
        uvm_analysis_imp_wrt #(fifo_transaction,coverage) wr_imp;
        uvm_analysis_imp_rdd #(fifo_transaction,coverage) rd_imp;

        fifo_transaction tx;

        int wclk_period;
        int rclk_period;
        int ratio_sampled;

        covergroup cg_clk_ratio;
                option.per_instance=1;
                clk_ratio_cp: coverpoint ratio_sampled {
                        bins wclk_much_faster = {[401:$]};
                        bins wclk_faster = {[200:400]};
                        bins near_equal = {[50:199]};
                        bins rclk_faster = {[25:49]};
                        bins rclk_much_faster = {[0:24]};
                }
        endgroup

        covergroup cg_fifo;
                option.per_instance=1;

                write_en:coverpoint tx.wr_en{
                        bins write={1};
                        bins zerow={0};
                }
                write_reset:coverpoint tx.wrst_n{
                        bins highw={1};
                        bins zerow={0};
                }
                write_data:coverpoint tx.wdata{
                        bins zero={32'h0};
                        bins low_range={[32'h0000_0001:32'h0000_FFFF]};
-- INSERT --
```

File Edit Tools Syntax Buffers Window Help

async_fifo_env.sv | coverage.sv | fifo_pkg.sv | gray_bind_coverage.sv | scoreboard.sv

```
        }
        write_data:coverpoint tx.wdata{
                bins zero={32'h0};
                bins low_range={[32'h0000_0001:32'h0000_FFFF]};
                bins mid_range={[32'h0001_0000:32'h7FFF_FFFF]};
                bins high_range={[32'h8000_0000:32'hFFFF_FFFF]};
        }
        walmost_transition:coverpoint tx.almostfull{
                bins almost_full={1};
                bins not_almost_full={0};
        }

        write_count:coverpoint tx.wfill_count{
                bins low={[0:3]};
                bins mid={[4:7]};
                bins high={[8:15]};

        }
        wfull_transition:coverpoint tx.wfull{
                bins full={1};
                bins not_full={0};
        }
        overflow:coverpoint tx.overflow_error{
                bins overflow={1};
                bins no_overflow={0};
        }
        read_reset:coverpoint tx.rrst_n{
                bins highr={1};
                bins zeror={0};
        }
        read_en:coverpoint tx.rd_en{
                bins readr={1};
                bins idle={0};
        }
        read_data:coverpoint tx.rdata{
                bins zeror={32'h0};
                bins low_ranger={[32'h0000_0001:32'h0000_FFFF]};
                bins mid_ranger={[32'h0001_0000:32'h7FFF_FFFF]};
                bins high_ranger={[32'h8000_0000:32'hFFFF_FFFF]};
```

-- INSERT --

Applications Places System

coverage.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/env) (2 of 5) - GVIM (on ip-10-

File Edit Tools Syntax Buffers Window Help

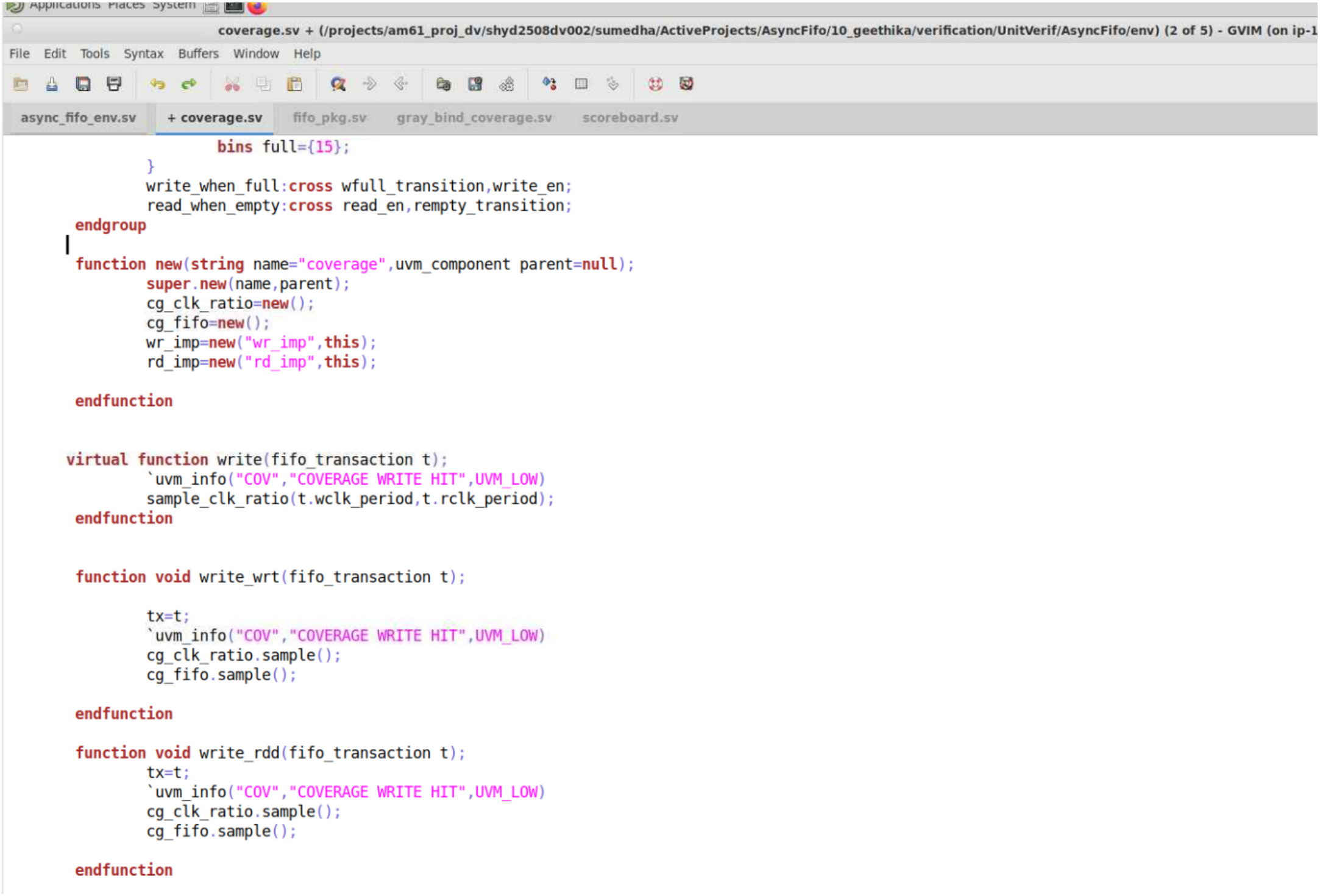
async_fifo_env.sv | coverage.sv | fifo_pkg.sv | gray_bind_coverage.sv | scoreboard.sv

```
                bins idle={0};
        }
        read_data:coverpoint tx.rdata{
                bins zeror={32'h0};
                bins low_ranger={[32'h0000_0001:32'h0000_FFFF]};
                bins mid_ranger={[32'h0001_0000:32'h7FFF_FFFF]};
                bins high_ranger={[32'h8000_0000:32'hFFFF_FFFF]};
        }
        rempty_transition:coverpoint tx.rempty{
                bins empty={0};
                bins not_empty={0};
        }

        ralmost_transition:coverpoint tx.almostempty{
                bins almost_empty={1};
                bins not_almost_empty={0};

        }

        read_count:coverpoint tx.rfill_count{
                bins lowr={[0:5]};
                bins midr={[6:10]};
                bins highr={[11:15]};
        }
        underflow:coverpoint tx.underflow_error{
                bins underflow={1};
                bins no_underflow={0};
        }
        cp_fifo_level:coverpoint tx.fifo_level{
                bins empty={0};
                bins low ={[1:5]};
                bins mid ={[6:10]};
                bins near_full={[11:14]};
                bins full={15};
        }
        write_when_full:cross wfull_transition,write_en;
        read_when_empty:cross read_en,rempty_transition;
endgroup
```


Applications Places System
coverage.sv + (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/env) (2 of 5) - GVIM (on ip-1
File Edit Tools Syntax Buffers Window Help
async_fifo_env.sv
+ coverage.sv
fifo_pkg.sv
gray_bind_coverage.sv
scoreboard.sv
bins full={15};
}
write_when_full:cross wfull_transition,write_en;
read_when_empty:cross read_en,rempty_transition;
endgroup
function new(string name="coverage",uvm_component parent=null);
super.new(name,parent);
cg_clk_ratio=new();
cg_fifo=new();
wr_imp=new("wr_imp",this);
rd_imp=new("rd_imp",this);
endfunction
virtual function write(fifo_transaction t);
`uvm_info("COV","COVERAGE WRITE HIT",UVM_LOW)
sample_clk_ratio(t.wclk_period,t.rclk_period);
endfunction
function void write_wrt(fifo_transaction t);
tx=t;
`uvm_info("COV","COVERAGE WRITE HIT",UVM_LOW)
cg_clk_ratio.sample();
cg_fifo.sample();
endfunction
function void write_rdd(fifo_transaction t);
tx=t;
`uvm_info("COV","COVERAGE WRITE HIT",UVM_LOW)
cg_clk_ratio.sample();
cg_fifo.sample();
endfunction

coverage.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/env) (2 of 5) - GVIM (on ip-10-115-24-57.cadence.com)

File Edit Tools Syntax Buffers Window Help

async_fifo_env.sv | coverage.sv | fifo_pkg.sv | gray_bind_coverage.sv | scoreboard.sv

```
      virtual function write(fifo_transaction t);
              `uvm_info("COV","COVERAGE WRITE HIT",UVM_LOW)
              sample_clk_ratio(t.wclk_period,t.rclk_period);
       endfunction


       function void write_wrt(fifo_transaction t);

              tx=t;
              `uvm_info("COV","COVERAGE WRITE HIT",UVM_LOW)
              cg_clk_ratio.sample();
              cg_fifo.sample();

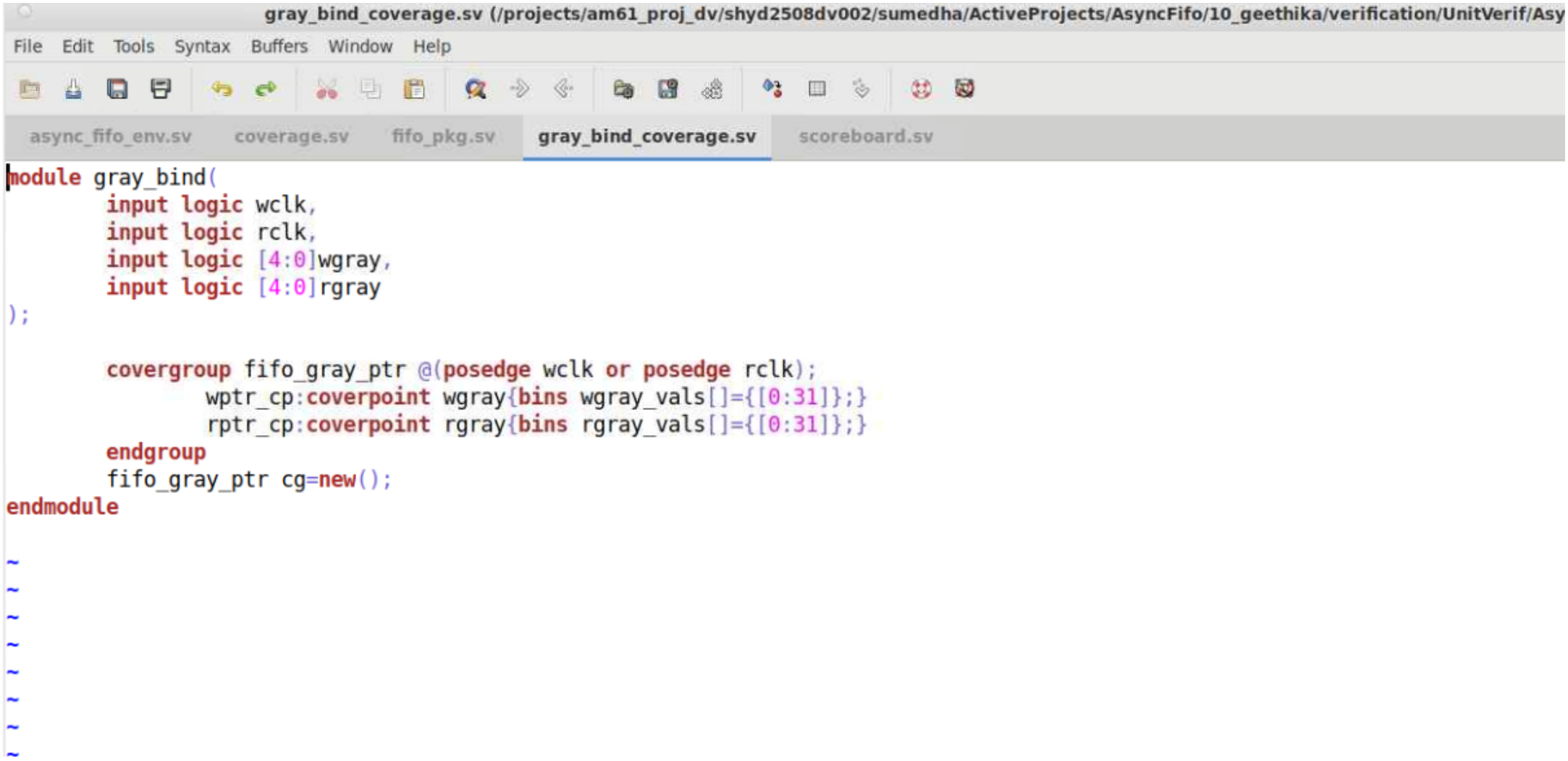
       endfunction

       function void write_rdd(fifo_transaction t);
              tx=t;
              `uvm_info("COV","COVERAGE WRITE HIT",UVM_LOW)
              cg_clk_ratio.sample();
              cg_fifo.sample();

       endfunction

       function void sample_clk_ratio(int w_per,int r_per);
              ratio_sampled=0;
              if(r_per==0) begin
                      `uvm_error("RATIO","RCLK_PERIOD IS 0")
                      return;
              end
              wclk_period=w_per;
              rclk_period=r_per;

              ratio_sampled=(wclk_period*100)/rclk_period;
              `uvm_info("RATIO",$sformatf("wclk=%0d rclk=%0d,clk_ratio_scaled=%0d",wclk_period,rclk_period,ratio_sampled),UVM_LOW)
              cg_clk_ratio.sample();
       endfunction
endclass
```


gray_bind_coverage.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/Asy
File Edit Tools Syntax Buffers Window Help
async_fifo_env.sv
coverage.sv
fifo_pkg.sv
gray_bind_coverage.sv
scoreboard.sv
module gray_bind(
input logic wclk,
input logic rclk,
input logic [4:0]wgray,
input logic [4:0]rgray
);
covergroup fifo_gray_ptr @(posedge wclk or posedge rclk);
wptr_cp:coverpoint wgray{bins wgray_vals[]={[0:31]};}
rptr_cp:coverpoint rgray{bins rgray_vals[]={[0:31]};}
endgroup
fifo_gray_ptr cg=new();
endmodule

assertion.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif

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```
module async_fifo_sva #(parameter DEPTH=16,parameter PTR_WIDTH=5)(
        input logic wrst_n,
        input logic rrst_n,
        input logic wr_en,
        input logic rd_en,
        input logic wclk,
        input logic rclk,
        input logic wfull,
        input logic rempty,
        input logic [31:0]wr_data,
        input logic [31:0]rd_data,
        input logic [4:0]wgray,
        input logic [4:0]rgray,
        input logic [3:0]waddr,
        input logic [3:0]raddr,
        input logic [4:0]wgray_sync);

        property p_data_integrity;
                @(posedge rclk)
                disable iff(!rrst_n)
                (rd_en && !rempty) |-> ##1 !$isunknown(rd_data);
        endproperty

        data_integrity:assert property(p_data_integrity)
                $display("test passed");
        else
                $error("DATA MISMATCH:FIFO order violated");

        property p_no_write_when_full;
                @(posedge wclk)
                disable iff(!wrst_n)
                (wfull && wr_en) |-> ##1 $stable(wgray);
        endproperty

        no_overflow:assert property(p_no_write_when_full)
                $display("test passed");
        else
                $error("invalid WRITE when full");

        property p_no_read_when_empty;
                @(posedge rclk)
                disable iff(!rrst_n)
                (rempty && rd_en) |-> ##1 $stable(rgray);
        endproperty

        no_underflow:assert property(p_no_read_when_empty)
                $display("test passed");
        else
                $error("invalid READ when empty");
```

assertion.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/assertions) -

File Edit Tools Syntax Buffers Window Help

```
                $error("invalid READ when empty");

        property p_gray_wptr;
                @(posedge wclk)
                disable iff(!wrst_n)
                (wgray !=$past(wgray)) |-> $onehot(wgray^$past(wgray));
        endproperty

        wgray_pointer:assert property(p_gray_wptr)
                $display("test passed");
        else
                $error("WGRAY NOT SINGLE BIT TRANSITION");

        property p_gray_rptr;
                @(posedge rclk)
                disable iff(!rrst_n)
                (rgray !=$past(rgray)) |-> $onehot(rgray^$past(rgray));
        endproperty

        rgray_pointer:assert property(p_gray_rptr)
                $display("test passed");
        else
                $error("RGRAY NOT SINGLE BIT TRANSITION");

        property p_full_flag;
                @(posedge wclk)
                disable iff(!wrst_n)
                (wfull && wr_en) |-> wfull;
        endproperty

        full_flag:assert property(p_full_flag)
                $display("test passed");
        else
                $error("FULL Flag not asserted correctly");

        property p_empty_flag;
                @(posedge rclk)
                disable iff(!rrst_n)
                (rgray==wgray_sync) |->  rempty;
        endproperty

        empty_flag:assert property(p_empty_flag)
                $display("test passed");
        else
                $error("EMPTY  Flag not asserted correctly");


endmodule
```